

Pierpaolo Vendittelli

AI Researcher · Computational Pathology · Medical Imaging

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EXPERIENCE

Postdoc Researcher

Dec. 2024 – Present

Prinses Maxima Centre for Pediatric Oncology

Utrecht, The Netherlands

- Developing DL and MIL pipelines for WSI analysis of Wilms tumor (anaplasia grading, clinical outcome prediction) across multiple collaborating hospitals.
- Translating model outputs into clinically interpretable biomarkers in collaboration with pathologists; supervising students and contributing to publications.

PhD Candidate – AI for Digital Pathology

Mar. 2021 – Jun. 2026 (exp.)

Radboud University Medical Center

Nijmegen, The Netherlands

- Developed multi-class U-Net for TSR quantification (PANCAIM, Horizon 2020): 368 patients / 4 datasets / 25 NL labs; Dice 0.75–0.86, TSR AUC 0.61 on TCGA-PAAD. Extended to multi-tissue segmentation for combining AI biomarkers (TSR, mitosis, stromal density) in PDAC survival analysis. Published in *PLOS ONE* (2024), *SPIE MI* (2022), *COMPAYL* (2025), *European Radiology* (2025), *Lancet Oncology* (2025).
- Teaching Assistant (*Image Analysis*, *Medical Imaging Analysis*, 4 editions); co-supervised Master's theses.

Machine Learning Engineer

Oct. 2020 – Feb. 2021

Everis Italy (NTT DATA)

Rome, Italy

- Built AI chatbots, annotated industrial data for object recognition, and developed live video translation pipelines for clients in banking and automotive.

Research Intern

Feb. 2020 – Sep. 2020

ViCOROB – NeuroImage Computing Group, University of Girona

Girona, Spain

- Benchmarked MRI synthesis architectures (ResU-Net, cGAN, CycleGAN) on WMH and BraTS 2018; ResU-Net achieved best SSIM/MSE under data-scarce conditions.

PUBLICATIONS & PROJECTS

Automatic quantification of tumor-stroma ratio as a prognostic biomarker for pancreatic cancer

PLOS ONE

doi:10.1371/journal.pone.0301969

May 2024

Multi-class U-Net; 368 patients / 4 datasets / 25 NL labs; Dice 0.75–0.86; TSR AUC 0.61 (6-month survival, TCGA-PAAD external cohort).

Automatic quantification of TSR as a prognostic marker for pancreatic cancer

MIDL 2023

openreview.net

Sep. 2023

Setting the research agenda for clinical AI in pancreatic adenocarcinoma imaging

doi:10.3390/cancers14143498

Cancers

Systematic review on AI for pancreatic cancer diagnosis and personalised treatment.

Jul. 2022

Automatic tumour segmentation in H&E-stained whole-slide images of the pancreas

SPIE MI 2022

doi:10.1117/12.2611542

Apr. 2022

End-to-end prognostication in pancreatic cancer by multimodal deep learning: a retrospective, multicenter study

European Radiology

2025

Artificial intelligence and radiologists in pancreatic cancer detection (PANORAMA): an international observational study

The Lancet Oncology

2025

Navigating the landscape of multimodal AI in medicine: a scoping review on technical challenges and clinical applications

Medical Image Analysis

2025

Combining AI-based Biomarkers for PDAC Survival

COMPAYL (MICCAI 2025 Workshop)

Sep. 2025

Weekly AI Paper-Triage Agent

github.com/PierpaoloV

Multi-agent LLM pipeline (OpenAI API, Python): weekly literature scan → summarisation → interest scoring → email digest.

Personal Project

2024 – Present

EDUCATION

Joint M.Sc. in Medical Imaging and Applications (MAIA)

University of Burgundy, University of Cassino, University of Girona

Sep. 2018 – Sep. 2020

France, Italy, Spain

- Grades: Bien (FR), 102/110 (IT), 8.6/10 (ES) • Erasmus Mundus Consortium Grant

B.Sc. in Informatics and Telecommunication Engineering

University of Cassino and Southern Lazio

Sep. 2012 – Sep. 2017

Cassino, Italy

- Grade: 94/110 • Erasmus+ Exchange, Fontys University of Applied Sciences, Eindhoven (2016)

SKILLS

Languages Python, C++, C, Bash, SQL

ML / Deep Learning PyTorch, scikit-learn, HuggingFace, Transformers, MONAI, Keras

MLOps & Infrastructure Docker, Git, SLURM / GPU clusters, OpenSlide, W&B

LLM & Agents OpenAI API, Anthropic API, prompt engineering, multi-agent orchestration